Roll No.: CL.AI.UMA152140

Amrita Vishwa Vidyapeetham

Amrita School of Artificial Intelligence, Faridabad

B.Tech Mid-Term Examinations – November 2025

Third Semester

Artificial Intelligence & Data Sciences

AI Optimization Techniques for Healthcare Resource Management

Duration: Two hours | Maximum: 50 Marks

Course Outcomes (COs):

CO	Course Outcomes
CO01	Understand healthcare systems, resources, and challenges, and formulate optimization problems.
CO02	Apply classical optimization techniques (LP, IP) and AI/ML models for hospital resource management.
CO03	Implement metaheuristic and evolutionary algorithms for healthcare scheduling, allocation, and routing.
CO04	Analyze and design AI-based decision support systems for patient flow, bed management, and emergency response.

Answer all questions

1) Formulate a linear programming problem for optimizing ambulance scheduling for amrita hospital (assume suitable numbers of ambulance and shifts). [5 marks] [CO02][BTL3 Level]
2) Compare traditional optimization techniques with AI/ML-based approaches in healthcare. [5 marks] [CO01][BTL2 Level]
3) A hospital faces fluctuating patient inflow across multiple departments. Propose a predictive modeling approach using machine learning to forecast patient numbers for the next month. Describe the features, model selection, evaluation metrics, and how the predictions can be integrated into staffing and bed allocation strategies. [5 marks] [CO01][BTL3 Level]
4) Describe the process of predictive modeling for forecasting equipment and drug demand in a hospital. [5 marks] [CO01][BTL2 Level]
5) A lab performs Blood Tests (T1) and X-Rays (T2). Each T1 takes 15 minutes, each T2 takes 30 minutes. Technician has 6 hours (360 min) per day. Max 16 T1 and 10 T2 can be done. Profit: 200 (T1), 500 (T2). Formulate to maximize profit. [5 marks] [CO02] [BT4Level]
6) Define healthcare resource management and explain its importance. [5 marks] [CO02] [BT2Level]

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7) A hospital has two operation theaters (OTs) and needs to schedule three types of surgeries: Cardiac, Orthopedic, and General. Each surgery type requires specific OT time and yields different revenue. Each OT is available for 8 hours per day. Formulate and solve (using graphical or simplex method) the LPP to maximize daily revenue. [8 marks] [CO02] [BT5Level]

Surgery Type	OT Time Required (hours)	Revenue per Surgery (₹)	Max Cases/Day
Cardiac	4	50,000	3
Orthopedic	3	30,000	4
General	2	20,000	5

8) A hospital needs to allocate 100 beds across three departments: General, ICU, and Maternity, to minimize patient waiting time while maximizing bed utilization.
   a) Formulate this as an optimization problem.
   b) Suggest whether linear or integer programming is suitable and justify your choice.
   c) Discuss how machine learning could further enhance this allocation process. [8 Marks] [CO01] [BTL5]

9) Differentiate between supervised and unsupervised learning by using a healthcare example. [4 Marks] [CO01] [BTL2]

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Course Outcome /Bloom's Taxonomy Level (BTL) Mark Distribution Table

CO	Marks	BTL	Marks
CO01	**27**	BTL 1	
CO02	**23**	BTL 2	**19**
CO03	-	BTL 3	**10**
CO04	-	BTL 4	**5**
CO05	-	BTL 5	**16**
CO06	-	BTL 6	-

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